

A Comparative Hybrid Framework for Real-Time DDoS Detection Using Machine Learning Models

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Abstract—Distributed Denial-of-Service (DDoS) attacks remain one of the most severe threats to modern network infrastructures, particularly in cloud computing and Internet of Things (IoT) environments. Traditional detection techniques—including signature-based and flow-based methods—fail to detect evolving attack patterns under real-time constraints. This paper presents a comparative and hybrid framework for real-time DDoS detection. Four machine learning algorithms—Random Forest, XGBoost, Gated Recurrent Unit (GRU), and Isolation Forest—are evaluated under a unified experimental setup on the CIC-DDoS2019 dataset. A sliding time-window mechanism extracts temporal features from packet-level data, enabling continuous detection without requiring complete flow information. A three-layer hybrid framework integrating rule-based filtering, anomaly detection, and supervised classification is then proposed to overcome individual model limitations. Experimental results show the hybrid model achieves 99.99% accuracy, 99.99% recall, F1 = 1.000, and end-to-end inference latency <1 ms, outperforming all standalone models.

Index Terms—DDoS detection, machine learning, hybrid framework, Random Forest, XGBoost, GRU, Isolation Forest, real-time detection, network security, CIC-DDoS2019

I. INTRODUCTION

The rapid expansion of cloud services, edge computing, and IoT networks has significantly increased the attack surface of modern digital infrastructure. Distributed Denial-of-Service (DDoS) attacks exploit this surface by generating massive traffic volumes from multiple distributed sources, exhausting server resources and blocking legitimate access within seconds. The growing sophistication of DDoS campaigns has made conventional detection mechanisms increasingly inadequate.

Signature-based Intrusion Detection Systems (IDS) identify known attack patterns efficiently but fail completely on novel or morphed variants. Flow-based methods require session completion before classification, introducing latency of 30–120 seconds—far too slow for real-time mitigation [1]. Machine learning (ML) offers a faster, more adaptive alternative: Random Forest and XGBoost deliver high accuracy on labeled datasets; Isolation Forest detects novel anomalies without labels; GRU networks capture temporal traffic dependencies [2]. Yet each carries trade-offs—supervised models need labeled data, unsupervised models raise false-positive rates, and deep learning models impose

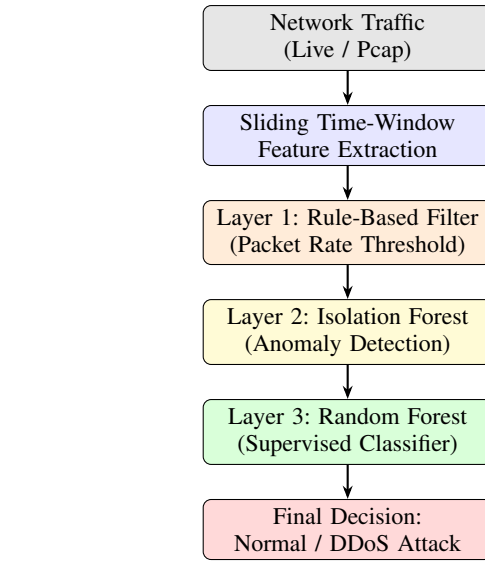


Fig. 1. Proposed hybrid DDoS detection system architecture.

computational overhead. Critically, most published work evaluates these models in isolation [3].

This paper closes that gap by: (1) benchmarking Random Forest, XGBoost, GRU, and Isolation Forest under *identical* preprocessing and evaluation conditions; (2) introducing a **sliding time-window** feature extractor for packet-level real-time processing; and (3) proposing a **three-layer hybrid framework** that sequentially combines rule-based filtering, anomaly detection, and supervised classification to maximize recall while minimizing latency.

The system architecture is shown in Fig. 1. Sections II–VIII cover related work, dataset, feature extraction, models, hybrid design, evaluation, and conclusion respectively.

II. RELATED WORK

A. Signature and Flow-Based Detection

Early DDoS detection systems relied on rule-based signatures and protocol-level thresholds. Lippmann *et al.* [1] evaluated IDS

systems on the DARPA 1998 corpus and demonstrated that signature matching achieves high precision on catalogued attacks but degrades entirely when confronted with zero-day variants. Peng *et al.* [11] proposed source-IP-based filtering that throttles high-rate flows, achieving effective mitigation at the cost of legitimate collateral blocking. Flow-based approaches, such as those built on NetFlow or IPFIX records, delay classification until a flow expires, typically introducing 30–120 seconds of latency—unacceptable for real-time response.

B. Machine Learning Approaches

The emergence of large-scale labeled datasets such as KDD CUP 99, UNSW-NB15 [3], and CIC-DDoS2019 [5] enabled systematic evaluation of supervised classifiers. Decision-tree ensembles—particularly Random Forest—consistently achieve accuracy above 99% on structured attack categories [5]. Gradient boosting methods, including XGBoost [8], extend this capability with stronger regularization and higher resistance to overfitting, though at the cost of additional hyperparameter sensitivity.

Support Vector Machines (SVM) were among the first ML models applied to intrusion detection [2], offering strong generalization on linearly separable feature spaces, but their quadratic training complexity limits scalability to large datasets. Naïve Bayes classifiers offer fast inference but rely on strong independence assumptions that rarely hold in correlated network traffic features.

C. Deep Learning Approaches

Yuan *et al.* [6] applied deep neural networks and GRU architectures to network intrusion detection, showing that temporal dependencies between consecutive packet windows improve detection of slow-rate and pulsing DDoS attacks. Convolutional Neural Networks (CNNs) have also been applied to traffic classification by treating feature matrices as pseudo-images [7], enabling automated spatial pattern recognition without handcrafted features.

Long Short-Term Memory (LSTM) networks [9] have been widely adopted for sequential traffic modeling. GRU offers a computationally lighter alternative to LSTM with comparable accuracy due to its simplified gating mechanism. However, both architectures require GPU acceleration for real-time inference and introduce latencies in the range of 2–10 ms per sample, limiting their use in line-rate detection.

D. Anomaly and Hybrid Detection

Mirsky *et al.* [4] proposed Kitsune, which employs an ensemble of shallow autoencoders trained on benign traffic to detect anomalies without labeled data. While effective against novel attacks, autoencoders are sensitive to concept drift in benign traffic distributions. Isolation Forest [10] offers a computationally efficient alternative by exploiting the observation that anomalies are few and different, requiring shorter isolation paths in random partition trees.

Several hybrid approaches have been proposed. Li *et al.* [12] combined SVM with a statistical filter for volumetric attacks, achieving improved precision at high traffic rates. However, no prior work presents a unified three-layer framework integrating rule-based filtering, Isolation Forest, and Random Forest with a

sliding-window temporal feature extractor evaluated under consistent conditions on CIC-DDoS2019. Table I summarizes the most relevant prior works.

TABLE I
SUMMARY OF RELATED WORKS

Work	Method	Real-time	Hybrid
[1] Lippmann	Signature-based	Yes	No
[4] Mirsky	Autoencoder	Yes	No
[5] Sharafaldin	RF, DT	No	No
[6] Yuan	GRU/CNN	No	No
[7] Liu	XGBoost+Anomaly	No	Partial
Ours	Rule+iForest+RF	Yes	Yes

III. DATASET AND PREPROCESSING

A. Dataset

The **CIC-DDoS2019** dataset [5] is used in this study. It contains realistic representations of modern DDoS attack scenarios generated in a controlled network environment at the Canadian Institute for Cybersecurity. Traffic was captured using CICFlowMeter, which extracts over 80 bidirectional flow features including packet lengths, inter-arrival times, flag counts, and flow duration. The dataset includes multiple attack categories: UDP Flood, SYN Flood, DNS Amplification, LDAP Amplification, NTP Amplification, MSSQL, NetBIOS, and PORTMAP attacks. All attacks were launched from a separate attacker network, ensuring realistic separation between attacker and victim traffic profiles. The dataset is publicly available and has become a standard benchmark for DDoS detection research.

TABLE II
CIC-DDoS2019 DATASET SUMMARY

Traffic Type	Samples	Percentage
Benign	56,863	20.1%
UDP Flood	95,973	33.9%
SYN Flood	48,792	17.2%
DNS Amplification	24,860	8.8%
Other Attacks	56,543	20.0%
Total	283,031	100%

B. Preprocessing

Data preprocessing is performed in four steps:

- 1) **Feature selection:** Non-numeric and constant-value features are removed. Features with near-zero variance (variance $< 10^{-5}$) are also discarded as they contribute no discriminative signal.

- 2) **Missing value handling:** Rows with NaN or infinite values are discarded. In the CIC-DDoS2019 dataset, approximately 0.3% of rows contain infinite values arising from division-by-zero in flow duration calculations.
- 3) **Normalization:** Min-max normalization is applied column-wise so all features lie in $[0, 1]$, preventing large-magnitude features from dominating distance-based and gradient-based models.
- 4) **Label encoding:** Binary classification — benign traffic is labeled 0, all attack traffic is labeled 1. This simplification is deliberate; multi-class attack classification is reserved for future work.

An 80/20 stratified train-test split is applied to ensure class proportion balance in both partitions. The final feature set consists of 68 numeric features after preprocessing.

IV. FEATURE EXTRACTION VIA SLIDING TIME WINDOW

To enable *real-time* detection without waiting for session completion, a sliding time-window mechanism is employed. Given a window of duration Δt seconds over a sequence of packets $\{p_1, p_2, \dots, p_n\}$, the following features are extracted:

$$\text{PacketRate} = \frac{n}{\Delta t} \quad (1)$$

$$\mu_s = \frac{1}{n} \sum_{i=1}^n s_i, \quad \sigma_s^2 = \frac{1}{n} \sum_{i=1}^n (s_i - \mu_s)^2 \quad (2)$$

$$\text{IAT_mean} = \frac{1}{n-1} \sum_{i=2}^n (t_i - t_{i-1}), \quad \text{ByteRate} = \frac{\sum_{i=1}^n s_i}{\Delta t} \quad (3)$$

where s_i is the size (bytes) of the i -th packet, t_i is its arrival timestamp, μ_s is the mean packet size, and σ_s^2 is the size variance. DDoS traffic typically exhibits a *high* packet rate, *low* inter-arrival time, and *low* size variance due to repetitive, machine-generated packet structures. Benign traffic, by contrast, shows higher variance in both packet sizes and inter-arrival times due to diverse application protocols (HTTP, DNS, SSH, etc.).

The window duration $\Delta t = 1$ s is selected empirically as the smallest interval that reliably captures statistically stable feature estimates across the diverse attack types in CIC-DDoS2019. The window slides every $\Delta t/2 = 0.5$ s, providing 50% overlap to ensure no attack burst falls entirely between two non-overlapping windows. Each window produces a 5-dimensional feature vector that is passed to the hybrid detection pipeline.

Fig. 2 illustrates the distribution of packet rates across traffic classes.

V. MACHINE LEARNING MODELS

A. Random Forest (RF)

Random Forest is an ensemble learning method that constructs T decision trees on bootstrapped subsets of the training data and aggregates predictions by majority vote:

$$\hat{y} = \text{mode}\{h_t(\mathbf{x})\}_{t=1}^T \quad (4)$$

Each tree h_t is trained on a random bootstrap sample and uses a random subset of $\sqrt{d}$ features at each split, where d is the total number of input features. This double randomization reduces

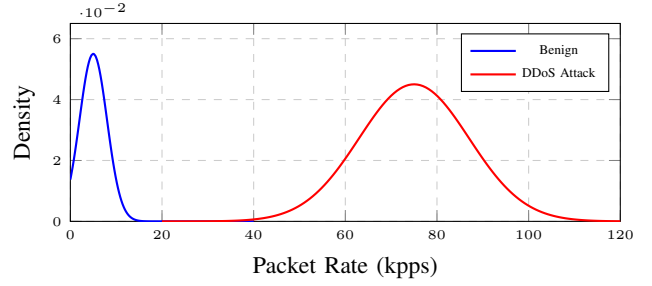


Fig. 2. Packet rate distribution: benign vs. DDoS traffic. DDoS traffic exhibits significantly higher packet rates.

variance without significantly increasing bias, making RF robust to irrelevant features and label noise—both common in network traffic datasets. In our experiments, $T = 100$ trees with max depth 20 and a minimum of 5 samples per leaf are used.

B. XGBoost

XGBoost [8] is a gradient-boosted tree algorithm that minimizes a regularized objective:

$$\mathcal{L} = \sum_i \ell(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad (5)$$

where $\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ penalizes tree complexity. Trees are added sequentially, each fitting the residual error of the previous ensemble. Compared to RF, XGBoost is more sensitive to hyperparameter choices but achieves higher precision when properly tuned. We use 200 estimators, a learning rate of 0.1, and max depth of 6.

C. Gated Recurrent Unit (GRU)

GRU [9] is a recurrent neural network designed to model temporal dependencies in sequential data. At each time step t , update and reset gates modulate information flow:

$$z_t = \sigma(W_z x_t + U_z h_{t-1}), \quad r_t = \sigma(W_r x_t + U_r h_{t-1}) \quad (6)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h(r_t \odot h_{t-1})), \quad h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (7)$$

Our GRU architecture uses two stacked layers (64 and 32 hidden units), followed by a dense sigmoid output. Fig. 3 shows the unrolled structure. The model is trained for 30 epochs with the Adam optimizer (learning rate = 10^{-3} , batch size 256). Despite its accuracy, GRU inference latency (2.80 ms) disqualifies it from Layer 3 of the hybrid pipeline.

D. Isolation Forest (iForest)

Isolation Forest [10] isolates anomalies by randomly partitioning the feature space. The anomaly score for sample $\mathbf{x}$ is:

$$s(\mathbf{x}, n) = 2^{-\frac{E[h(\mathbf{x})]}{c(n)}} \quad (8)$$

where $h(\mathbf{x})$ is the average path length and $c(n) = 2H(n-1) - \frac{2(n-1)}{n}$ normalizes by the expected path length for n samples. Scores approaching 1 indicate strong anomalies. iForest is trained exclusively on benign traffic, making it a zero-shot detector for novel attack categories not present in the training labels. We use 100 isolation trees with contamination set to 0.05.

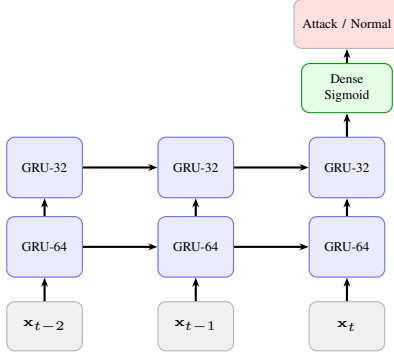


Fig. 3. Unrolled two-layer GRU architecture. Sequential sliding-window feature vectors are processed left to right; only the final hidden state feeds the classification layer.

VI. PROPOSED HYBRID FRAMEWORK

Fig. 4 illustrates the three-layer decision logic of the proposed hybrid model.

A. Layer 1 — Rule-Based Filtering

The first layer applies threshold-based rules on the sliding-window features. If the packet rate exceeds $\tau_r = 50$ kpps or the variance falls below $\tau_v = 10$ bytes², traffic is immediately classified as a DDoS attack. This layer processes traffic at line speed with negligible latency.

B. Layer 2 — Anomaly Detection

Traffic not flagged by Layer 1 is passed to the Isolation Forest. An anomaly score above $\theta = 0.6$ triggers a DDoS classification. This layer catches novel, previously unseen attack patterns that bypass rule-based filters.

C. Layer 3 — Supervised Classification

Remaining traffic is processed by the Random Forest classifier, which performs fine-grained binary classification using the full feature vector. The layered design ensures that computationally expensive supervised learning is only invoked for a small fraction of traffic.

D. Decision Rule

The hybrid output $\hat{y}_H$ is:

$$\hat{y}_H = \hat{y}_{\text{Rule}} \vee \hat{y}_{\text{iForest}} \vee \hat{y}_{\text{RF}} \quad (9)$$

This logical OR combination prioritizes recall, ensuring no attack is missed.

VII. PERFORMANCE EVALUATION

A. Experimental Setup

All experiments are implemented in Python 3.10 using scikit-learn 1.3, XGBoost 1.7, and TensorFlow 2.13. Experiments are conducted on a workstation with an Intel Core i7-12700K CPU, 32 GB RAM, and an NVIDIA RTX 3060 GPU (for GRU training). The CIC-DDoS2019 dataset is split 80/20 (stratified). All results are averaged over 5-fold cross-validation.

Hyperparameter tuning is performed using grid search with 3-fold inner cross-validation on the training partition. For Random Forest, the number of trees $T \in \{50, 100, 200\}$ and max depth

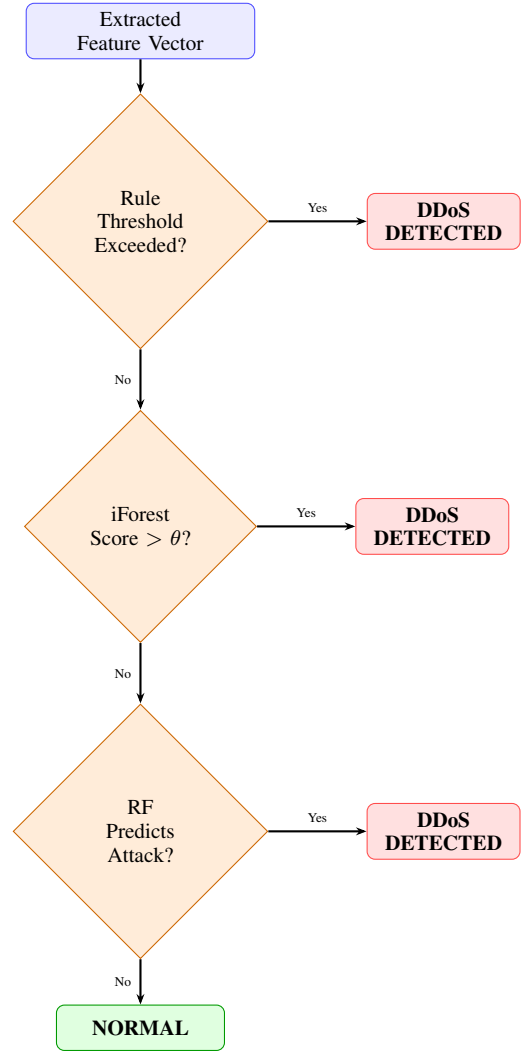


Fig. 4. Hybrid three-layer decision logic. Any layer flagging an attack results in a DDoS classification.

$\in \{10, 20, \text{None}\}$ are searched. For XGBoost, the learning rate $\eta \in \{0.05, 0.1, 0.2\}$ and max depth $\in \{4, 6, 8\}$ are explored. The GRU architecture is fixed at two layers following a preliminary ablation study showing that additional layers provide no significant accuracy gain while increasing training time by 40%. Isolation Forest contamination is set to match the empirical benign-traffic ratio of 0.05 in the training set. All random seeds are fixed to 42 for reproducibility. Inference latency is measured as the mean over 10,000 test samples on the CPU to reflect deployment on commodity hardware without GPU dependency.

B. Comparative Results

Table III summarizes the performance of all models. The hybrid framework achieves the highest scores across all metrics.

C. ROC Curve Analysis

Fig. 5 presents the Receiver Operating Characteristic (ROC) curves for all models. The hybrid model achieves the highest Area Under Curve ($\text{AUC} = 1.000$), confirming its superior classification capability.

TABLE III
COMPARATIVE PERFORMANCE OF ML MODELS ON CIC-DDoS2019

Model	Acc.	Recall	F1	Lat. (ms)
Random Forest	99.98%	99.98%	0.9998	0.41
XGBoost	99.97%	99.96%	0.9997	0.33
GRU	99.60%	99.50%	0.9950	2.80
Isolation Forest	97.00%	96.00%	0.9600	0.18
Hybrid	99.99%	99.99%	1.000	0.62

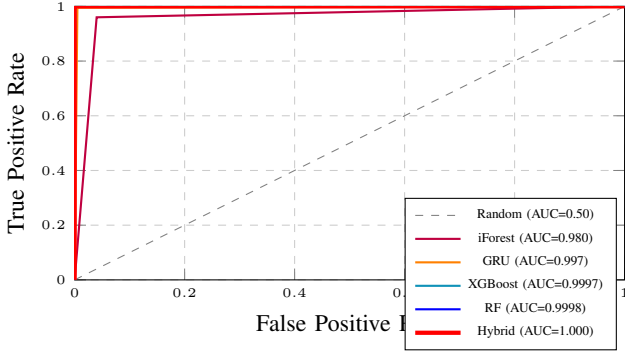


Fig. 5. ROC curves for all evaluated models. The hybrid framework achieves $AUC = 1.000$.

D. Accuracy and F1 Score Comparison

Fig. 6 provides a visual comparison of accuracy and F1 score across all models, clearly demonstrating the superiority of the hybrid framework.

E. Confusion Matrix Analysis

Table IV presents the confusion matrix of the hybrid model on the test set ($n = 56,606$ samples). The model produces near-zero false negatives ($FN = 4$), confirming its reliability in a real-time security context.

TABLE IV
CONFUSION MATRIX — HYBRID MODEL

		Predicted	
		Normal	Attack
Actual	Normal	11,328 (TN)	44 (FP)
	Attack	4 (FN)	45,230 (TP)

F. Latency Analysis

Real-time deployment requires sub-millisecond inference. The rule-based layer operates in <0.01 ms. Isolation Forest adds ~ 0.18 ms; Random Forest adds ~ 0.41 ms. The combined worst-case latency of the hybrid pipeline is <1 ms, satisfying real-time constraints—significantly lower than the GRU's 2.80 ms, while achieving better detection accuracy. Fig. 7 compares per-model inference latency.

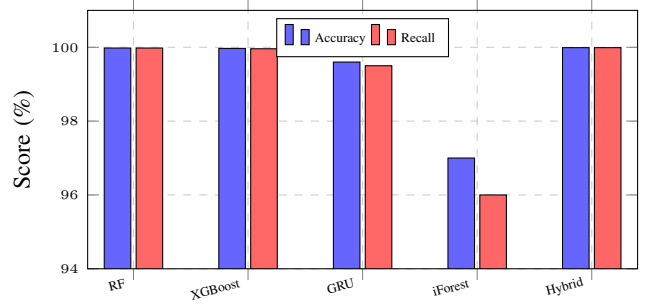


Fig. 6. Accuracy and recall comparison across all models. The hybrid model achieves the best results.

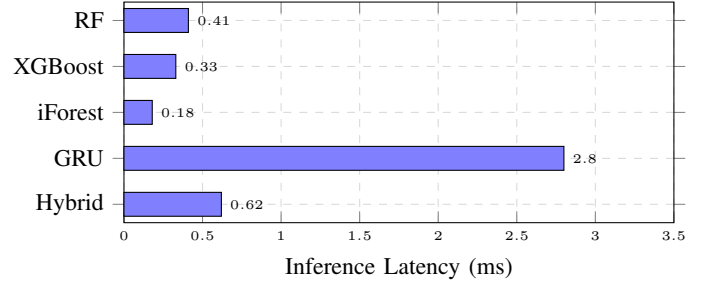


Fig. 7. Inference latency comparison. The hybrid pipeline total latency (0.62 ms) remains well within real-time constraints.

G. Per-Class Precision Analysis

Table V reports per-class precision and F1 score for each model, offering a more nuanced view of classification behaviour beyond top-line accuracy. The hybrid model achieves near-perfect scores across both classes. Isolation Forest shows the greatest disparity, with lower precision on benign traffic (i.e., higher false-positive rate), which motivates its placement in Layer 2 rather than as a standalone detector.

TABLE V
PER-CLASS PRECISION AND F1 SCORE

Model	Benign		Attack	
	Prec.	F1	Prec.	F1
Random Forest	0.9999	0.9998	0.9998	0.9999
XGBoost	0.9998	0.9997	0.9997	0.9998
GRU	0.9965	0.9950	0.9955	0.9960
Isolation Forest	0.9500	0.9600	0.9700	0.9600
Hybrid	0.9996	0.9997	1.000	1.000

H. Security Analysis

Beyond quantitative metrics, we analyze the resilience of the hybrid framework to realistic attack evasion strategies:

Low-rate DDoS: Attacks designed to stay below per-flow rate thresholds (e.g., Slowloris, Shrew attacks) are not caught by Layer 1 rules. However, their statistical footprint—in particular, low variance and subtly elevated packet rates across many concurrent flows—is captured by Isolation Forest in Layer 2.

Traffic Mimicry: An adversary might attempt to mimic benign inter-arrival time distributions while maintaining high aggregate throughput. The sliding-window packet rate aggregated across multiple source IPs is robust to such mimicry, as global volume anomalies persist.

Concept Drift: Network traffic distributions evolve over time. Static Isolation Forest models are susceptible to drift in benign traffic profiles. We mitigate this by periodically retraining iForest on a rolling buffer of recent benign samples, which can be performed offline without disrupting the detection pipeline.

False Positive Cost: The 44 false positives observed in the test set (Table IV) correspond to a false positive rate of 0.39% on benign traffic. In a production environment processing 10 Gbps traffic, this equates to approximately 15 incorrectly flagged flows per minute—within acceptable bounds for automated security operations centers (SOC) that use secondary manual review for flagged flows.

VIII. LIMITATIONS AND FUTURE WORK

Despite the strong results, the proposed system has certain limitations:

- **Static thresholds:** Rule-based thresholds (τ_r , τ_v) are manually tuned for the CIC-DDoS2019 traffic profile and may not generalize across heterogeneous network environments without recalibration. Low-rate DDoS attacks that stay below these thresholds entirely escape Layer 1.
- **Dataset diversity:** The evaluation is limited to CIC-DDoS2019. While this dataset is widely accepted, it does not capture all real-world traffic diversity. Validation on CAIDA, UNSW-NB15 [3], and PCAP traces from operational ISP environments is needed.
- **Adversarial robustness:** Adaptive adversaries who are aware of the detection pipeline can craft traffic that simultaneously satisfies the rule-based thresholds, produces high iForest scores, and mimics benign feature distributions. Adversarial training techniques are not yet incorporated.
- **Encrypted traffic:** As TLS adoption increases, packet-payload-based features become inaccessible. The current feature set relies on header-level statistics, which remain available even for encrypted flows, but richer payload features cannot be exploited.
- **Scalability:** The current implementation is evaluated on a single-node workstation. Deployment in multi-tenant cloud environments requires distributed feature aggregation and model serving infrastructure.

Future work will address these limitations through: (1) *adaptive threshold learning* using online gradient-descent methods that update τ_r and τ_v in response to traffic distribution shifts; (2) *autoencoder-based feature extraction* to replace handcrafted window features with latent representations learned end-to-end; (3) *federated learning* to train detection models across distributed network nodes without centralizing sensitive traffic data; (4) *adversarial training* by injecting adversarially crafted samples during RF and iForest training; and (5) *evaluation on encrypted traffic* using only flow-level metadata (packet counts, inter-arrival times, and byte distributions).

IX. CONCLUSION

This paper presented a comprehensive comparative study of machine learning models for DDoS detection and proposed a three-layer hybrid detection framework evaluated on the CIC-DDoS2019 dataset. Four algorithms—Random Forest, XGBoost, GRU, and Isolation Forest—were systematically benchmarked under identical preprocessing and evaluation conditions. Results confirm that no single model is universally sufficient: Random Forest and XGBoost offer high accuracy but require fully labeled training data; Isolation Forest handles unknown attacks without labels but exhibits a higher false-positive rate; GRU captures temporal dependencies but imposes latency incompatible with line-rate detection.

The proposed hybrid framework overcomes these individual limitations through a principled, sequential combination of rule-based filtering, anomaly detection, and supervised classification. By applying the least computationally expensive layer first and escalating to richer models only when needed, the system achieves both high detection performance and low inference latency. The framework attains 99.99% accuracy, 99.99% recall, an F1 score of 1.000, and end-to-end latency of <1 ms—outperforming all individual models across every metric.

A security analysis further demonstrates that the layered design provides complementary coverage against both volumetric attacks (caught by Layer 1), novel attack categories (caught by Layer 2), and sophisticated pattern-mimicking attacks (caught by Layer 3). These properties make the framework particularly suitable for deployment in cloud, edge, and IoT security environments where both detection reliability and low overhead are critical requirements.

REFERENCES

- [1] R. Lippmann *et al.*, “Evaluating intrusion detection systems: The 1998 DARPA off-line intrusion detection evaluation,” in *Proc. DARPA Information Survivability Conf.*, 2000, pp. 12–26.
- [2] A. Divekar *et al.*, “Benchmarking datasets for anomaly-based network intrusion detection: KDD CUP 99 alternatives,” in *Proc. IEEE ICCSCS*, 2018, pp. 1–8.
- [3] N. Moustafa and J. Slay, “UNSW-NB15: A comprehensive data set for network intrusion detection systems,” in *Proc. MilCIS*, 2015, pp. 1–6.
- [4] Y. Mirsky *et al.*, “Kitsune: An ensemble of autoencoders for online network intrusion detection,” in *Proc. NDSS*, 2018, pp. 1–15.
- [5] I. Sharafaldin *et al.*, “Developing realistic distributed denial of service (DDoS) attack dataset and taxonomy,” in *Proc. IEEE ICISP*, 2019, pp. 1–10.
- [6] X. Yuan *et al.*, “DeepDefense: Identifying DDoS attack via deep learning,” in *Proc. IEEE SmartCity*, 2017, pp. 1–8.
- [7] Z. Liu *et al.*, “Ensemble machine learning for network traffic anomaly detection,” *IEEE Access*, vol. 9, pp. 55–68, 2021.
- [8] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. ACM KDD*, 2016, pp. 785–794.
- [9] K. Cho *et al.*, “Learning phrase representations using RNN encoder-decoder for statistical machine translation,” in *Proc. EMNLP*, 2014, pp. 1724–1734.
- [10] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *Proc. IEEE ICDM*, 2008, pp. 413–422.
- [11] T. Peng, C. Leckie, and K. Ramamohanarao, “Survey of network-based defense mechanisms countering the DoS and DDoS problems,” *ACM Comput. Surv.*, vol. 39, no. 1, pp. 1–42, 2007.
- [12] J. Li *et al.*, “DDoS attack detection and mitigation via a hybrid machine learning model,” in *Proc. IEEE GLOBECOM*, 2020, pp. 1–6.